

SIMATS Engineering

SIMATS Online Programming Lab — Python Programming (Mock Interview)

Course Name	Python Programming
Course Code	CSA0804
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Register Number	1972260035
Topic	CO3-AT3-MOCK INTERVIEW
Question	Q1 — Python Lists vs. C Arrays (10 Marks)

Q1. [10 Marks]

What is a list in Python? How is it different from an array in C? Give an example of a heterogeneous list.

Answer

What is a list in Python?

A list in Python is a built-in, ordered, and mutable data structure that can hold a collection of items. Lists are created using square brackets, e.g. `[1, 2, 3]`, and support operations such as indexing, slicing, appending, inserting, and removing elements. Unlike some other collection types, a list can grow or shrink dynamically at runtime and does not require its size to be declared in advance.

How is a Python list different from a C array?

- **Data type:** a C array can only store elements of one single, fixed data type (e.g. all ints or all floats), whereas a Python list can store elements of different data types together in the same list.
- **Size:** a C array has a fixed size that must be declared at creation and cannot change afterward, while a Python list is dynamic — it automatically resizes as elements are appended or removed.
- **Memory layout:** a C array stores its elements in one contiguous block of memory of a known fixed element size, while a Python list stores references (pointers) to objects, which may be scattered in memory and can be of varying size.
- **Built-in operations:** Python lists come with many built-in methods (append, remove, sort, etc.), whereas C arrays offer no such built-in methods — operations like inserting or resizing must be coded manually.
- **Type safety:** C arrays are strictly typed at compile time, catching type mismatches at compile time; Python lists are dynamically typed, so type-related issues typically surface only at runtime.

Example of a heterogeneous list

Because Python lists are not restricted to one data type, a single list can freely mix integers, strings, floats, booleans, and even other lists — something a C array cannot do.

Python Program

```
"""
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Demonstrates that a Python list can hold multiple data types
```

```
at once (heterogeneous), unlike a C array which is restricted
to a single, fixed data type.
"""

# A heterogeneous list: mixes int, str, float, bool, and even
# another list, all in one Python list
student_record = [101, "Ravi", 8.75, True, [90, 85, 88]]

print("Heterogeneous list:", student_record)
for item in student_record:
    print(f"Value: {item!r:<20} Type: {type(item).__name__}")
```

Sample Output

```
Heterogeneous list: [101, 'Ravi', 8.75, True, [90, 85, 88]]
Value: 101                Type: int
Value: 'Ravi'             Type: str
Value: 8.75               Type: float
Value: True               Type: bool
Value: [90, 85, 88]       Type: list
```

Conclusion

A Python list is a dynamic, built-in collection that can grow, shrink, and hold mixed data types, while a C array is a fixed-size, single-type block of contiguous memory with no built-in resizing or utility methods. The `student_record` example demonstrates this flexibility directly: a single list simultaneously holds an int (101), a str ("Ravi"), a float (8.75), a bool (True), and even a nested list ([90, 85, 88]) — a mix that a C array could never represent without resorting to structs or unions.